LECTURE 1

Def'n A <u>vector</u> is an element of the set

$$\mathbb{R}^n = \left\{ \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} : x_1, \dots, x_n \in \mathbb{R} \right\}$$

Notation

$$\vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \quad \vec{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

Ex

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \in \mathbb{R}^2, \quad \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \in \mathbb{R}^3$$

Note

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \neq \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

Proposition <u>Operations on Vectors</u>

① Addition $\vec{x} + \vec{y} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix}$

② Scalar multiplication If $c \in \mathbb{R}$, $\vec{x} \in \mathbb{R}^n$, then

$$c\vec{x} = c \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} cx_1 \\ \vdots \\ cx_n \end{bmatrix}$$

③ Subtraction $\vec{x} - \vec{y} = \begin{bmatrix} x_1 - y_1 \\ \vdots \\ x_n - y_n \end{bmatrix}$, i.e. $\vec{x} + (-1)\vec{y}$

④ Vector multiplication $\vec{x} \cdot \vec{y} = \begin{bmatrix} x_1 y_1 \\ \vdots \\ x_n y_n \end{bmatrix}$

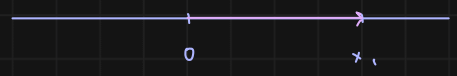
<u>NOT DOT / CROSS PRODUCT</u>

vector, notation, operations on vectors

CONTINUED

Proposition — Geometry of vectors

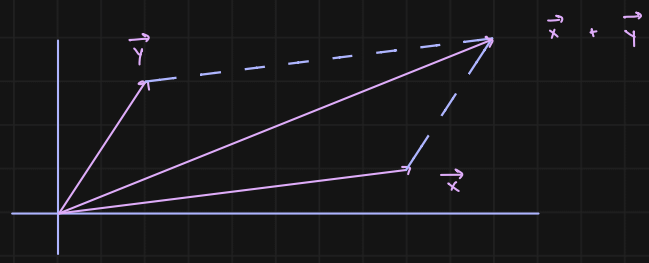
$\mathbb{R}^n$. $n = 1 \Rightarrow \vec{x} = \begin{bmatrix} x_1 \end{bmatrix} \in \mathbb{R}^1$



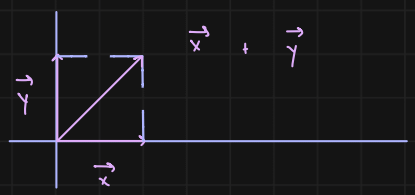
$n = 2 \Rightarrow \vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \in \mathbb{R}^2$

Proposition — Geometric interpretation of additions

Theorem — Parallelogram Law



Ex — Add $\vec{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\vec{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$



We have $\vec{x} + \vec{y} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$= \begin{bmatrix} 1 + 0 \\ 0 + 1 \end{bmatrix}$

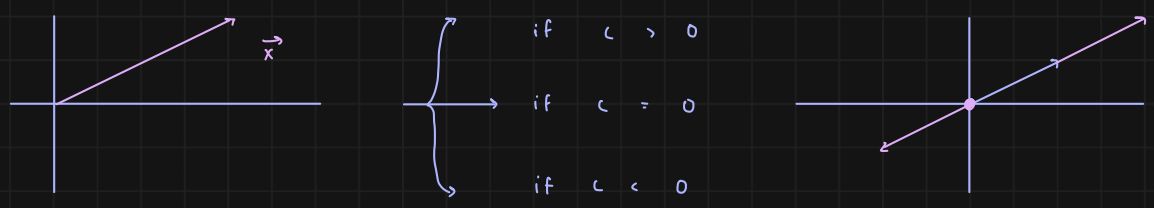
$= \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

vector geometry, geometric interpretation of addition, parallelogram law

CONTINUED

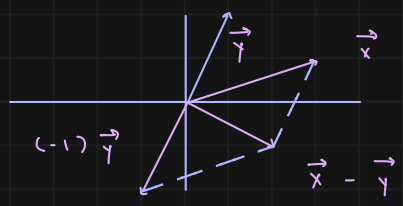
Proposition — <u>Geometric interpretation of scaling</u>

Let $c \in \mathbb{R}$ and $\vec{x} \in \mathbb{R}^2$



Proposition — <u>Geometric interpretation of subtraction</u>

Let $\vec{x}, \vec{y} \in \mathbb{R}^2$. Then $\vec{x} - \vec{y} = \vec{x} + (-1)\vec{y}$



Def'n — The <u>zero vector</u> in $\mathbb{R}^n$ is the vector $\vec{0} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$

Ex — $\vec{0}_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $\vec{0}_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Proposition — <u>Properties of addition and scalar multiplication</u>

1. commutativity $\forall\ \vec{x}, \vec{y} \in \mathbb{R}^n$,
   $\vec{x} + \vec{y} = \vec{y} + \vec{x}$
2. associativity $\forall\ \vec{x}, \vec{y}, \vec{z} \in \mathbb{R}^n$,
   $(\vec{x} + \vec{y}) + \vec{z} = \vec{x} + (\vec{y} + \vec{z})$
3. $\forall\ s, t \in \mathbb{R}$, $\vec{x} \in \mathbb{R}^n$,
   $(s + t)\vec{x} = s\vec{x} + t\vec{x}$
4. $\forall\ s \in \mathbb{R}$, $\vec{x}, \vec{y} \in \mathbb{R}^n$,
   $s(\vec{x} + \vec{y}) = s\vec{x} + s\vec{y}$

   transitivity
5. cancellation $\forall\ c \in \mathbb{R}$, $\vec{x} \in \mathbb{R}^n$,
   if $c\vec{x} = \vec{0}$, then $c = 0$, $\vec{x} = 0$, or both

geometric interpretation of scaling and subtraction, zero vector, properties of addition and scalar multiplication